

1.

Thank you Mr/Ms chair.

I'm Ryo Yamatomi.

I'm from Shonan Institute of Technology in Japan.

Today, I will introduce our recent research, entitled the

“Generative Model of Suitable Meme Sentences for Images Using AutoEncoder”.

2.

This is the outline of my presentation.

First, I will talk about the introduction.

3.

Recent years, neural networks have been used in various fields.

However, there are few researches using neural networks in joking.

Laughing is an important factor for health.

But, due to Covid-19, laughter decreased from peoples.

So, it can be said that there are demand for laughter by AI without fear of infection.

Therefore, this research implements an AI that generates funny sentences for images.

In this research, the word “Meme” is a funny sentence for an image.

4.

Conventional method used pre-trained image recognition model as an image encoder,
and the model sometime generate memes unsuitable for images due to domain divergence.

The proposed method uses AutoEncoder to deal with domain divergence.

Our proposed method trained better than conventional method.

Also, the proposed method has flexibility for the model structure.

5.

A meme is one of cybercultures spreading and attracting on social network services.

These images are examples of memes.

As these images show, a meme is a pair of an image and a funny sentence suitable for the image.

In Japan, Meme is popularly known as Ogiri.

6.

Bokete is a website allowing users to post Ohgiri for images.

Ohgiri is a Japanese meme.

Now, almost 100 million memes and almost 7.3 million images have been posted on the website.

As a reminder, all Bokete images have no label.

These memes are posted on Bokete.

These English sentences were translated from Japanese by DeepL.

7.

Next, I will talk about the conventional research.

8.

“Neural Joking Machine”, NJM is similar to our research.

This research was scraping Bokete website to collect pairs of images and memes.

9.

The collected data was used for training an image caption generative model.

In this model, images are used as input and Meme sentences are used as teacher signals.

This figure is the structure of the model.

As this figure shows, ResNet152, pre-trained on ImageNet, was used as an image encoder.

This research freezed the parameters of the ResNet152 while training, this method is called Transfer Learning.

Transfer Learning is often used in image caption generative models because it reduces the number of parameters during training.

The fun of the memes generated by the meme generator were evaluated by questionnaire.

10.

NJM has a problem.

NJM sometime generate memes unsuitable for images.

These memes are examples of it.

As you can see, these memes are unsuitable for images.

We considered domain divergence is the cause of this problem.

11.

Domain divergence is the difference of domain between pre-trained images and actual task images.

Domain divergence is a serious problem that causes a decrease in learning accuracy.

These natural images are from ImageNet.

These images are from Bokete. As these images show, Bokete images have specific-domain, such as Animations and Comics and etc.

It can be imagined that a model trained on ImageNet would not be able to extract the features of Bokete-specific images.

Therefore, the goal of this research is to deal with the domain divergence and to generate memes with higher accuracy.

12.

Next, I will talk about our proposed method.

13.

In order to deal with domain divergence, we need an image encoder suitable for Bokete-specific images.

So, we proposed a new method using the encoder of the AutoEncoder, pre-trained on Bokete-specific images, as the image encoder of an image caption generative model.

This figure is the structure of the proposed method.

14.

AutoEncoder trains to restore its input, thus, it doesn't require the label for data.

It is suitable for unlabeled Bokete images.

Also, the structure of a AutoEncoder is defined freely.

15.

The first step of the proposed method, training AutoEncoder using Bokete images.

16.

The second step, freeze the parameters of the encoder and train the image caption generative model.

17.

We call this method the “Generative Model of Suitable Meme Sentences for Images using AutoEncoder” as GUMI-AE.

18.

This table shows the comparison between GUMI-AE and NJM.

The conventional method, NJM used transfer learning.

Transfer learning has a high risk of domain divergence.

Because the image encoder had not trained on Bokete-specific images.

Also, pre-trained image encoders don't have flexibility for their structure.

In contrast, the proposed method, GUMI-AE requires the AutoEncoder to be trained in advance.

However, since the image encoder is trained on Bokete-specific images, domain divergence doesn't occur.

In addition, the structure of AutoEncoder can be defined freely, so users can use a model that is appropriate for their computer.

19.

Next, I will talk about experiments.

20.

As in the conventional research, we created a dataset by scraping Bokete website for training a meme generator.

The details of created data are shown as this slide.

21.

In this experiment, training AutoEncoder.

The output dimension of the encoder is changed to several patterns to verify its effect.

The details of experiment are shown as this slide.

22.

This table is the results of this experiment.

Each row are the results of the output dimension of the encoder.

As this table shows, increasing the output dimension of the encoder resulted in better performance.

We consider the increased output dimension of the encoder to have also increased the expressive power.

The following experiment uses this encoder.

23.

In this experiment, training two types of meme generators under the same conditions and data.

This figure is the basic structure of the model.

The difference of them is the image encoder.

First is the proposed method, GUMI-AE, using the encoder of the autoencoder trained previous experiment.

Second is the conventional method, NJM, using ResNet152 pre-trained on ImageNet.

The details of experiment are shown as this slide.

24.

These figures are training loss and training accuracy.

The y-axes are training loss and training accuracy, and the x-axes are training epochs.

In both figures, the red lines are GUMI-AE, and blue lines are NJM.

These figures show, GUMI-AE achieves .775 lower training loss and 12% higher accuracy than NJM for the same epoch.

It can be said, GUMI-AE resolved the domain divergence and trained with high accuracy.

These meme generators trained under same conditions and data without the image encoder.

Thus, the reason of this difference is the performance of the image encoder.

25.

These memes were generated by GUMI-AE.

These images are created by pasting AI-generated Meme sentence onto the image.

These memes are translated from Japanese by DeepL.

These images were not used for training.

As these images show, GUMI-AE is able to generate a meme suitable for the images.

26.

Compare the memes generated by GUMI-AE and NJM.

In these examples, when natural images are input, both methods generated funny memes suitable for the image.

In this Meme, the dog is appealing to be taken for a walk, which is suitable for the image.

27.

In this Meme, the word "captured" is used for a starfish that has been brought ashore, which is also suitable for the image.

This meme mistakes a starfish for a jellyfish, which turned out funny.

28.

In these examples, when not natural images be input, GUMI-AE generated memes suitable for the images.

But NJM generated memes not suitable for the images.

In this meme, the text "Where's Wally?" might be generated because the coloring of the character in the image looks like the character in "Where's Wally?".

29.

This meme contains unsuitable words for images such as "tadpoles" and "pod" and is not funny!

30.

Finally, we evaluated the fun of the memes generated by the proposed method.

As in the conventional research, we evaluated with questionnaire if a meme is fun.

In this questionnaire, humans evaluate the memes generated by GUMI-AE, NJM, caption AI, and Humans.

The details of questionnaire are shown as this slide.

31.

This figure is the result of this questionnaire.

The y-axis is the fun-level, and the x-axis is the percentage of votes.

The reds are GUMI-AE, the blues are NJM, greens are the caption AI, and yellows are humans.

This figure shows GUMI-AE received more positive reactions than NJM.

However, still humans received more positive reactions than GUMI-AE.

This figure also shows many of the memes generated by the AI are rated as boring.

This may be because of the roulette selection for word selection.

Since roulette selection has randomness in its selection, there is a possibility of selecting grammatically incorrect words.

This might generate grammatically incorrect sentences and they were rated as boring.

32.

Last is conclusion.

We proposed a new method for an AI generate memes for images using AutoEncoder, called GUMI-AE.

GUMI-AE dealt with domain divergence compared to the conventional method.

Experiments showed the implemented meme generator generates funny memes suitable for the images.

According to the questionnaire, GUMI-AE generates funnier memes than NJM.

33.

In the future work, we will implement a meme generator using advance network, such as Gated Recurrent Units or Transformer to generate funnier memes than humans.

Also, implement the AI evaluate if a meme is fun for quantitated evaluation.

In addition, knowledge discovery from generative process of a meme generator to understand how to memes are created.

This is the end of the presentation.

Thank you for listening!

Any question??

34.

The hyperparameters were defined by my experience.

I will do it in future work.

35.

The used image caption generative model consists of CNN image encoder and LSTM sentence decoder.

In training image caption generative model, inputs are image and sentence up to the middle, output is probability distribution of recommendations of following word of input sentence.

Human choice the word from the probability distribution by argmax or roulette selection.

In argmax, choose the word which probability distribution indicates the maximum.

36.

The selected words are combined with sentence up to the middle,

and input the sentence is input recursively into the model.

Finery, sentences are generated.

37.

38.

This table is inputs and outputs of AutoEncoder.

These images were used training, and these images are output images corresponding the images its input.

As you can see, the AutoEncoder is able to restore its input if the training images are input.

These images were not used training, and these images are output images corresponding the images its input.

Some parts of these images were not restored, but we can say that they have been restored sufficiently.

39.

This figure is experimental results with a lightweight model.

In this experiment, the encoder output dimensions are 512 and 2048.

This experiment shows that the AutoEncoder encoder achieved lower training loss and higher training accuracy than the ResNet152 encoder.

40.

In roulette selection, human select the word randomly according to output probability distribution.

Roulette selection randomly selects words to generate a variety of sentences, but this may result in the selection of grammatically incorrect words.

41.

There are 2 types of metrics for sentence generation.

First is like BLUE or Perplexity.

These methods are word matching between the test and generated sentences.

In an ordinary image caption generation task would work well because the test and training data are similar.

In meme generation task would not well because the test and training data are not similar.

Second is BERT Score.

This method is mean matching between the test and generated sentences.

This method depends on the BERT vocabulary.

Bokete data has many words that are not in BERT's vocabulary, so it does not work.

In the conventional research did not use these metrics for Meme evaluation for the same reasons as ours.